Assignment 2

In this assignment, you will use a **Large Language Model (LLM)** to assist you in designing a visualization.

The purpose of the assignment is **not** to see whether an LLM can automatically produce a good visualization. Instead, you will **prompt, evaluate, critique, and guide an LLM toward an effective visualization design**.

You are expected to apply the visualization principles and concepts covered in the course.

Dataset: You will work with the dataset provided on moodle.

The data contains 6 attributes for 193 countries. These data, based on 2017 measures collected by the World Bank. The attributes are:

- **country**: The name of the country
- **region**: The world region (Africa, Americas, Asia, or Europe)
- **income**: Income per person based on the gross domestic product per person, adjusted for differences in purchasing power (in international dollars, fixed for 2017 prices)
- **life_exp**: Average number of years a newborn child would live if current mortality patterns were to stay the same
- **co2**: Carbon dioxide emissions per person from the burning of fossil fuels (metric tonnes of CO2 per person)
- **co2_change**: Indicator of whether the carbon dioxide emissions per person from the burning of fossil fuels has increased or decreased since 2007 (10 year span)
- **population**: Population of the country, in millions

For each question below report which LLM provider and model you used:

Model: OpenAI, ChatGPT default model
OR
Model: Google, Gemini 3.1 Pro

Question 1 — Generate (30%)

Use an **LLM to create an initial set of visualizations**.

Provide the LLM with the dataset and ask it to recommend and create **2 visualizations**.

Submit: Your prompt, the LLM's response, and the resulting visualizations.

Question 2 — Critique (40%)

Critically evaluate the visualizations produced in Question 1 using the visualization principles covered in the course.

For each visualization:

- Identify a **specific task or analytical goal** for which it is well-suited.
- Identify and explain any **design flaws or opportunities for improvement**.

Submit: Your critique and proposed improvements.

Question 3 — Improve & Explain (30%)

Use the LLM to iteratively refine your visualizations based on your critique from Question 2. You may conduct multiple rounds of refinement, and you are free to accept or reject the LLM's suggestions.

For each round, document:

- The **specific improvement** you intended to make and the **prompt** used to generate it.
- The resulting **visualization**.
- A brief explanation of **what changed and why**.

For the **final round** explain the overall changes and why the design is better.

You may accept or reject the LLM's suggestions.

Submit: Your prompts, resulting visualizations, explanations of the changes, and final design justification.